

Power BI Next Process Execution Blueprint

Complete Technical Guide: Post-Modeling, Advanced DAX, Report Layout & Deployment

DAX Analytics

RLS Security

Performance Optimization

Service Deployment

1. EXECUTIVE PHASE OVERVIEW

Following the successful establishment of the Star Schema relationships between DIM_product, BOM_data, FACT_Listings, FACT_Warranty_Listings, and DIM_Date, this guide details the next critical technical phases required to build, secure, optimize, and publish an enterprise-grade reporting solution.

2. ENTERPRISE DAX METRIC LAYER

Create a dedicated **_Measures Table** in Power BI Desktop to centralize all analytical logic. Avoid creating implicit measures in visual fields.

A. Core Volume & Coverage Metrics

```
Total Active Listings = COUNTROWS(FACT_Listings)
```

```
Total Warranty Claims = COUNTROWS(FACT_Warranty_Listings)
```

```
High Confidence Listing Ratio =  
DIVIDE(  
    CALCULATE(  
        COUNTROWS(FACT_Listings),  
        FACT_Listings[confidence_level] = "High"  
    ),  
    [Total Active Listings],  
    0  
)
```

B. Advanced Quality & Failure Rates

```
Claim Rate Per Listing =  
DIVIDE(  
    [Total Warranty Claims],  
    [Total Active Listings],  
    0  
)
```

```
Component Failure Rate =  
DIVIDE(  
    COUNTROWS(FACT_Warranty_Listings),  
    CALCULATE(  
        SUM(BOM_data[qty_per_unit]),  
        CROSSFILTER(FACT_Warranty_Listings[parent_sku], DIM_product[parent_sku], Both)  
    ),  
    0  
)
```

```
0
)
```

C. Time Intelligence Measures

```
Claims YTD =
TOTALYTD(
    [Total Warranty Claims],
    DIM_Date[Date]
)
```

```
Claims MoM Growth =
VAR CurrentMonth = [Total Warranty Claims]
VAR PreviousMonth = CALCULATE([Total Warranty Claims], DATEADD(DIM_Date[Date], -1, MONTH))
RETURN
DIVIDE(CurrentMonth - PreviousMonth, PreviousMonth, 0)
```

3. REPORT VIEW ARCHITECTURE & LAYOUT BLUEPRINT

Structure your report workbook into three dedicated thematic page layouts tailored for executive, operational, and granular investigative audiences.

Page Name	Target Visuals	Slicers & Filters	Primary KPI Focus
1. Executive Dashboard	KPI Cards, Line Chart (Scrapes vs Claims), Donut (Confidence Split)	Year, Quarter, Product Category	Active Listings, Total Claims, High-Confidence Ratio %
2. BOM Quality Analytics	Stacked Bar (Failures by Component Type), Matrix (SKU vs MTBF)	Component Type, Supplier, Damage Flag	Failure Rate %, MTBF Hours, Component Unit Cost
3. Warranty Detail Grid	Interactive Data Matrix with Conditional Formatting (Red/Green)	Failure Reason, Location, Within Warranty Flag	Claim ID, Scrape Date, Failure Date, Total Cost

4. PERFORMANCE OPTIMIZATION & SECURITY GOVERNANCE

A. Performance Optimization Workflow

- **DAX Studio Audit:** Connect DAX Studio to analyze Storage Engine (SE) vs Formula Engine (FE) CPU time. Optimize measures with high FE usage.
- **Column Cardinality Reduction:** Remove high-cardinality metadata (e.g., raw URLs or free-text descriptions) from fact tables to optimize VertiPaq compression.
- **Performance Analyzer:** Run Performance Analyzer in Power BI Desktop to ensure visual render times remain under 1.2 seconds.

B. Row-Level Security (RLS) Configuration

In Power BI Desktop, navigate to **Modeling → Manage Roles** and define dynamic security filters based on product manufacturing sites or product categories.

```
[Role: Regional Manager]
DIM_product[manufacturing_site] = USERPRINCIPALNAME()
```

5. POWER BI SERVICE DEPLOYMENT & REFRESH AUTOMATION

1. **Publishing:** Navigate to **Home** → **Publish** and select the target workspace (e.g., *Operations Analytics Production*).
2. **Gateway & Credentials:** In Power BI Service, navigate to Dataset Settings → Data Source Credentials to configure cloud/on-premise gateway authentication.
3. **Scheduled Refresh:** Configure automated daily scheduled refreshes (e.g., 06:00 AM and 06:00 PM) to ensure data currency.
4. **App Distribution:** Create a Power BI App workspace package for end-user distribution with audience-level visibility controls.